

Homework Problems. *Bonus Problems*

Definition. A *group homomorphism* is a function $f : G_1 \rightarrow G_2$ between two groups that preserves the group operation; i.e., $f(ab) = \underbrace{f(a)f(b)}_{\text{operation of } G_2}$.

ex. The mapping $f : \mathbb{Z} \rightarrow \mathbb{Z}_n$ where $\mathbb{Z}$ and $\mathbb{Z}_n$ are groups under addition and modular addition respectively. Since $f(a+b) = (a+b) \bmod n = a \bmod n + b \bmod n$.

Definition. Let G be a group and H a subset of G . For any $a \in G$, the set $aH = \{ah | h \in H\}$ is called the *left coset of H in G containing a* . $Ha = \{ha | h \in H\}$ is called the *right coset of H in G containing a* .

Lemma. For H a subgroup of G and $a, b \in G$, the following properties of cosets hold:

1. $a \in aH$.
2. $aH = H$ if and only if $a \in H$. (parts 1 and 2 must be done together)
3. $aH = bH$ or $aH \cap bH = \emptyset$.
4. $|aH| = |Ha| = |bH| = |H|$.

1. **Lagrange's Theorem:** $|H| \mid |G|$. If G is a finite group and H is a subgroup of G , then $|H|$ divides $|G|$. (also $\frac{|G|}{|H|}$ is the number of distinct left cosets in G).

2. Prove that if $|G| \leq 4$ then G is abelian. Then prove that if $|G| < 6$ then G is abelian.

3. Let $f : G_1 \rightarrow G_2$ be a group homomorphism and let $H \leq G_2$. Prove that $f^{-1}(H) = \{g \in G_1 | f(g) \in H\}$ is a subgroup of G_1 .